Kyle McGovern, PhD
Bioinformatics Scientist

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Summary

Bioinformatics scientist with 8+ years of experience developing rigorous and reproducible statistical and machine learning methods for next-generation sequencing (NGS) data analysis. Expert in building production-ready pipelines and open source tools for analyzing RNA-seq, single-cell, longitudinal, spatial transcriptomic, and 16S rRNA-seq data. Extensive experience with gene editing, transcriptomics, and microbiome studies. Deep expertise in Bayesian modeling, generalized linear modeling, Gaussian processes, Bayesian neural networks, and partially identified models. Demonstrated success leading tool development, developing custom CRISPR and AAV analysis pipelines, and contributing to patented therapeutic platforms.

Research Experience

Pennsylvania State University | *Graduate Researcher* | State College, PA | 2020-2025

- Improved accuracy of machine learning and inference on sequence count data (e.g., 16S rRNA-seq) through Bayesian Partially Identified Models (BPIMs)
- Developed a Bayesian hierarchical framework for mixed-effects modeling, correlation analysis, and machine learning (e.g., Bayesian Neural Networks) of sequence count data, addressing compositionality and reducing false positives by up to 80% with minimal loss of power
- Developed novel statistical theory for improved hypothesis testing in differential expression/abundance analyses
- Developed a bioinformatics and statistical pipeline for predicting chromosomal sex from contaminant host reads in metagenomic sequencing data

Sangamo Therapeutics | *Bioinformatics Engineer II* | Richmond, CA | 2018-2020

- Built a patented pipeline for tissue-targeted AAV protein engineering, automating AAV viral efficiency ranking, sequence homology clustering, and uncertainty quantification
- Developed Nanopore and Illumina sequencing pipeline for barcoding AAV variants for multiplexed engineering of AAV capsids

Caribou Biosciences | *Bioinformatics Programmer* | Berkeley, CA | 2015-2017

- Developed SITE-Seq: a custom CRISPR off-target detection tool in Python for an in vitro assay, leveraging statistical read clustering and motif analysis to assess genome editing specificity and actionable target identification
- Developed a Flask and D3.js web tool to visualize NHEJ, MMEJ, and (micro) homology directed repair patterns from amplicon sequencing data in gene editing experiments

Education

PhD, Bioinformatics & Genomics | Penn State, PA | 2020-2025

Dissertation Title: Statistical and Computational Methods for Addressing the Scale Limitation of Sequence Count Data
Advisor: Dr. Justin Silverman

BS, Bioinformatics | UC Santa Cruz, CA | 2011 - 2015

Publications

McGovern, KM, Silverman, J. **Scale Reliant Mixed Effects Models Enhance Microbiome Data Analysis**. *bioRxiv*, 2025. doi: 10.1101/2025.08.05.668734 | jsilve24/ALDEx3.

McGovern, KM, Silverman, J. **Replacing normalizations with interval assumptions enhances DE/DA**. *BMC Bioinformatics*, 2025. doi: 10.1186/s12859-025-06177-2 | Silverman-Lab/INDExA.

McGovern, KM, Nixon, M., Silverman, J. **Addressing erroneous scale assumptions in gene set enrichment**. *PLoS Computational Biology*, 2023. doi: 10.1371/journal.pcbi.1011659.

Nixon, M., McGovern, KM, et al. **Scale Reliant Inference**. *arXiv*, 2022. doi: 10.48550/arXiv.2201.03616.

Amrani, N., Gao, X., . . ., McGovern, KM, et al. **NmeCas9 is an intrinsically high-fidelity genome-editing platform**. *Genome Biology*, 2018. doi: 10.1186/s13059-018-1591-1.

Open Source Software

jsilve24/**ALDEx3** | 2025
R package for scale-reliant differential abundance including mixed effects modeling
Role: Co-author, Contributor

Silverman-Lab/**INDExA** | 2024
Interval Null Differential Expression and Abundance Analysis
Role: Author

Patents

ENGINEERING AAV | 2022
Patent # 18630996
David Ojala, McGovern, KM.

Skills

Languages
R, Python, Javascript, Java, C++

Software Packages
sk-learn, pandas, numpy, TensorFlow, SciPy, matplotlib, seaborn, ggplot2, tidyverse, Scanpy, Plotly, flask, django, D3.js, unittest, Pytest

Bioinformatics Tools
BLAST, samtools, Snakemake, BWA, FastQC, DADA2, QIIME, MetaPhlAn

Computational Tools
Linux, HPC, git, LaTeX, airflow, docker

Databases
MySQL, PostgreSQL, MongoDB

Statistics & Machine Learning Partially-identified Bayesian Models, Bayesian neural networks, conformal prediction, causal inference, random forests, xgboost, Gaussian & Student's T-processes, mixed effects models, hypothesis testing, PCA, SVMs, t-SNE, UMAP, sandwich estimators, MCMC, naive Bayes, Gaussian mixture models, generalized linear models (GLMs)